MTH 446: Lecture # 4

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Lecture Span

1. Roots of complex numbers

Simple multiplication example

In general, evaluating if $z_1 z_2 = z_3$, we can either (1) distribute and evalaute element-by-element or (2) check that the modulus of the RHS = LHS and that the arguments of both are the same.

Need to review what values of cos correspond to θ values.

To obtain the argument of a complex number, calculate

$$z = |z| \left(\frac{z}{|z|} \right) \tag{1}$$

Here, then the complex number $z/|z|$ corresponds to some $e^{i\theta}$. We can find this by solving

$$\cos\theta = x, \quad \sin\theta = y \tag{2}$$

For $z/|z| = x + iy$. Note, the argument of a product is the sum of the arguments of each product term.

Theorem 0.1. *De Moisre's formula states that*

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta \tag{3}$$

Proof: use $e^{i\theta} = \cos\theta + i\sin\theta$.

We can use De Moisre's formula to derive useful trigonometric identities, and matching the imaginary part with the other imaginary part, and etc.

Roots of complex numbers

Let $z = re^{i\theta}$, then we want to find

$$w^n = z \tag{4}$$

Here, then

$$w^n = re^{i\theta} \tag{5}$$

Let $w = \rho e^{i\phi}$, then

$$\rho^n = r, \quad n\phi = \theta + 2\pi k,\ k \in \mathbb{Z} \tag{6}$$

Clearly, $\rho = r^{1/n}$, and

$$\phi = \theta/n + 2\pi k/n \tag{7}$$

Therefore,

$$z^{1/n} = \{r^{1/n} \exp\left(i(\frac{\theta}{n} + \frac{2\pi k}{n})\right) : \quad k = 0, \dots, n-1\} \tag{8}$$

There are many values of this number.

Definition 0.2. *Let $z \in \mathbb{C}$, then let $\theta = \arg(z)$ (principal value of $\arg(z)$), then*

$$r^{1/n} e^{i \arg(z)/n} \tag{9}$$

This is the <u>principal root</u>, with the notation of $\sqrt[n]{z}$ (as opposed to $z^{1/n}$).

Suppose $z \neq 0$ and $z = re^{i\theta}$, then

$$z^{1/2} = \{c_0, c_1\} \iff \{\sqrt{r}e^{i\theta/2}, \sqrt{r}e^{i(\theta/2 + 2\pi/2) \iff \sqrt{r}} e^{i\theta/2} e^{i\pi} = -\sqrt{r}e^{i\theta/2}\} \tag{10}$$

As another example,

$$1^{1/2} = \{1, -1\} \tag{11}$$

However, $\sqrt{1}$ (principal root) is 1.

Example

Evaluate $(-16)^{1/4}$ and find the principal value. Here, $\pi = \arg(-16)$. Then,

$$c_k = 2e^{(\pi/4 + 2\pi k/4)i} \tag{12}$$

Calcuate

$$c_0 = 2\left(\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) \tag{13}$$

$$c_1 = 2\left(-\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right) \tag{14}$$

$$c_2 = 2\left(-\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right) \tag{15}$$

$$c_3 = 2\left(\frac{\sqrt{2}}{2} - i\frac{\sqrt{2}}{2}\right) \tag{16}$$

Here, then the principal root is $c_0 = \sqrt{2} + i\sqrt{2}$ because $\pi = \arg(z)$. **This will probably be on the midterm.**